LAN JIN

Computational Plasma Physics | Electron Emission | Semiconductor Equipment R&D

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PROFESSIONAL SUMMARY

Ph.D. candidate in Nuclear Engineering and Radiological Sciences at the University of Michigan, expected summer 2027. Research spans circuit-coupled gas discharge, electron emission, and scientific ML. Co-author of PASCHEN-1D and first author in Physical Review Research and Journal of Applied Physics. Prior TFT-LCD mask-design experience connects modeling with manufacturing; co-inventor on a pending U.S. patent and two granted Chinese utility-model patents.

CORE COMPETENCIES

Modeling: exact TDSE, electron emission, Poisson/drift-diffusion, fluid and particle-in-cell plasma simulation, and circuit-coupled gas discharge

Fusion & Plasma: preionized-plasma startup, early ionization, plasma formation and edges, sheaths, CCP/ICP, plasma-surface interaction

Scientific ML: physics-guided modeling, surrogate and interpretable models, regression, ensembles, neural networks; plasma and atmospheric-science applications

Engineering: simulation automation, HPC workflows, technical communication, IP development, cross-functional manufacturing support

PROFESSIONAL EXPERIENCE

Summer Research — AI in the Nuclear Sector **Jun 22, 2026 - Aug 29, 2026**

University of Michigan | Nuclear Engineering and Radiological Sciences | Supervisor: Prof. Aditi Verma

- Investigated AI applications in nuclear safety, security, and safeguards, emphasizing governance, regulation, and the historical development of AI in the nuclear sector.
- Contributed to survey design and development for a nuclear-sector AI observatory to collect data on AI use, oversight, and emerging trends.

Graduate Research Assistant **Jan 2025 - Present**

University of Michigan | Nuclear Engineering and Radiological Sciences

- Contributed to PASCHEN-1D, a 1D fluid plasma solver incorporating photoemission, thermionic emission, secondary-electron emission, and RLC circuit coupling for gas-breakdown and Paschen-scaling studies.
- Investigate ultrafast photoemission from metals using exact TDSE methods; analyze energy-resolved spectra, quantum efficiency, pulse-shape effects, and carrier-envelope-phase control.
- Build ML surrogates and interpretable models with Scikit-learn, TensorFlow, and Keras to accelerate simulation campaigns and identify nonlinear parameter dependencies.
- Published first-author research showing orders-of-magnitude improvement in photoemission quantum efficiency by shortening laser pulses at fixed low intensity; mapped spectral, photon-order, and carrier-envelope-phase behavior.
- Co-authored two 2026 atmospheric-science papers on physics-guided machine learning for spatiotemporal turbulence-dissipation reconstruction and Doppler-lidar mixing-layer-height retrieval.
- Contribute preionized-plasma modeling to the U-M team on Helion Energy's HERCULES fusion program; collaborate across theory, simulation, and laboratory work on startup physics.

Graduate Research Assistant **Jan 2022 - Dec 2024**

Michigan State University | Electrical and Computer Engineering

- Developed an exact-quantum simulation of direct density modulation from an RF cold cathode under combined RF and optical excitation; characterized current-pulse amplitude, width, shape, harmonics, and modulation depth.
- Co-invented an optically gated field-emission cathode for fast, flexible electron-beam control; U.S. nonprovisional patent application filed Aug. 9, 2024 and pending.

- Presented technical results to plasma and vacuum-electronics audiences at APS-DPP, IEEE ICOPS, IVEC/IVESC, IVNC, GEC, and MIPSE; received the 2024 MIPSE Fellowship.

TV Mask Design Engineer **Apr 2018 - Sep 2019**

Chuzhou HKC Photoelectric Technology Co., Ltd. | TFT-LCD volume manufacturing

- Designed photolithography masks for mass-produced TFT-LCD panels and supported process, yield, and product-engineering teams during volume manufacturing.
- Co-invented two display-panel designs granted Chinese Utility Model Patents in 2020.

Co-trained Research Student **Aug 2015 - Jan 2017**

Institute of Physics, Chinese Academy of Sciences | Ultracold Atoms and Polar Molecules Group

- Helped develop a high-performance 2D+ magneto-optical trap for 40K and designed free-space electro-optical modulators for high-speed optical switching.

EDUCATION

Ph.D., Nuclear Engineering and Radiological Sciences **Expected summer 2027**

University of Michigan | Major/Research Area: Plasmas & Nuclear Fusion

M.S., Electrical and Computer Engineering **2022 - 2024**

Michigan State University

M.S., Physical Electronics **2014 - 2017**

Yunnan University

B.S., Fundamental Science of Mathematics and Physics **2010 - 2014**

Yunnan University

TECHNICAL SKILLS

Programming: Python (NumPy, SciPy, Pandas, Matplotlib), MATLAB, C/C++, Mathematica

Simulation: exact TDSE; Poisson/drift-diffusion; finite-difference, finite-volume, and spectral methods; WarpX, EDIPIC, PIC, fluid and hybrid plasma models

Machine Learning: Scikit-learn, TensorFlow, Keras; regression, ensembles, neural networks, surrogates, feature attribution

Research Computing: Linux/HPC, Git, SLURM, LaTeX, reproducible scientific workflows

SELECTED PUBLICATIONS & INTELLECTUAL PROPERTY

1. Iqbal, A., Heri, Y., Wang, B., Jin, L., Faisal, M. A., and Zhang, P. "PASCHEN-1D: A one-dimensional fluid plasma solver with multi-mechanism surface emission and flexible external circuit coupling." Computer Physics Communications 329, 110404 (2026). DOI: 10.1016/j.cpc.2026.110404
2. **Jin, L.**, and Zhang, P. "Enhanced quantum efficiency in photoemission driven by few-cycle optical fields." *Physical Review Research* 8, 013035 (2026). DOI
3. **Jin, L.**, Zhou, Y., and Zhang, P. "Direct density modulation of photo-assisted field emission from an RF cold cathode." *Journal of Applied Physics* 134, 074904 (2023). DOI
4. Chu, Y., et al. (including Jin, L.) Surface-constrained reconstruction of spatiotemporal turbulence dissipation in the convective boundary layer using physics-guided machine learning. Journal of Geophysical Research: Atmospheres 131, e2026JD047285 (2026). DOI: 10.1029/2026JD047285
5. Chu, Y., et al. (including Jin, L.) A Physically-Guided and Stage-Wise Machine Learning Framework for Mixing Layer Height Retrieval Based on Doppler Lidar Observations. Advances in Atmospheric Sciences (2026). DOI: 10.1007/s00376-026-6100-y
6. Zhang, P., **Jin, L.**, and Zhou, Y. "Optically Gated Field Emission Cathodes." U.S. patent application; nonprovisional filed Aug. 9, 2024; patent pending.
7. **Jin, L.**, co-inventor. Two granted Chinese utility-model patents for display-panel designs: CN201920865304.2 and CN201920864409.2 (2020).